Detekcia rohov a záujmových bodov

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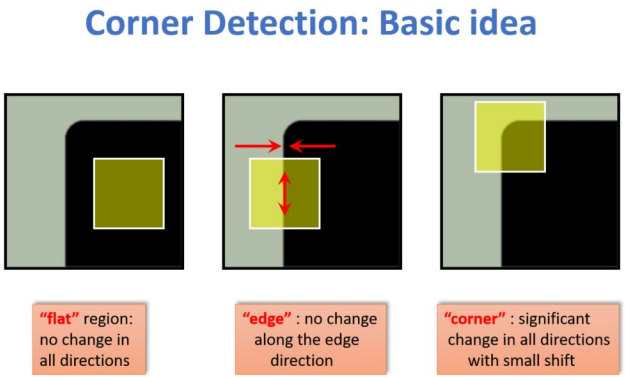
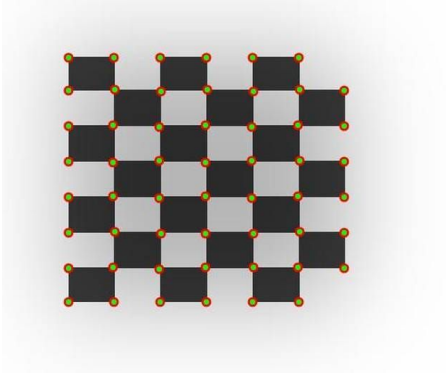
Ciele

- Harrisov detektor rohov
- Výpočet matice druhých momentov
- Odozvu rohu a prahovanie
- Porovnať s OpenCV implementáciou

Princíp Harrisovho detektora

- roh = bod kde sa intenzita mení vo viacerých smeroch súčasne
- posúvanie okna po obraze - sledovanie zmeny obsahu
  - plocha: žiadna zmena,
  - hrana: zmena v 1 smere,
  - roh: zmena vo všetkých smeroch





Výpočet gradientov (Sobelov operátor)

- Sobelove konvolučné jadrá:
  - Sobel X – horizontálny gradient (zmeny zľava doprava)
  - Sobel Y – vertikálny gradient (zmeny zhora nadol)
- Manuálna konvolúcia cez sliding_window_view a np.einsum
- Výstup sú gradienty Gx, Gy a magnitúda

```
[-1,-2,-1]
[0, 0, 0]
[1, 2, 1]
```

```
[-1, 0, 1]
[-2, 0, 2]
[-1, 0, 1]
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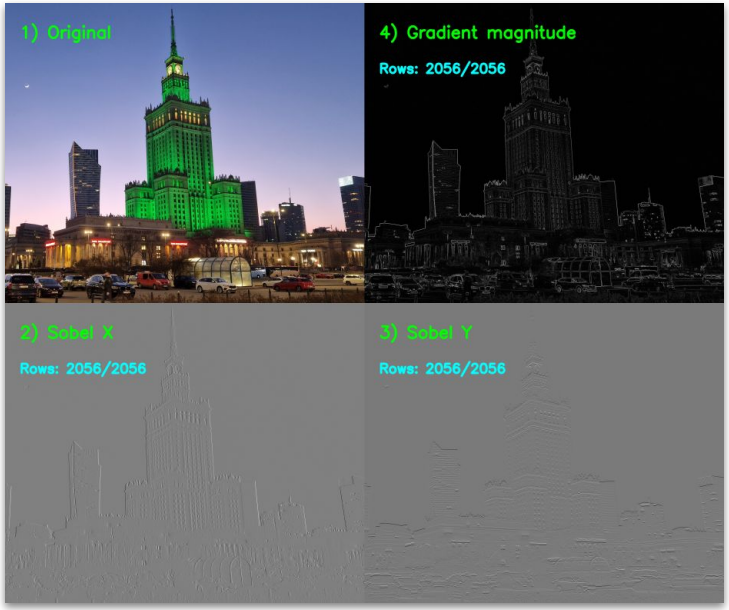
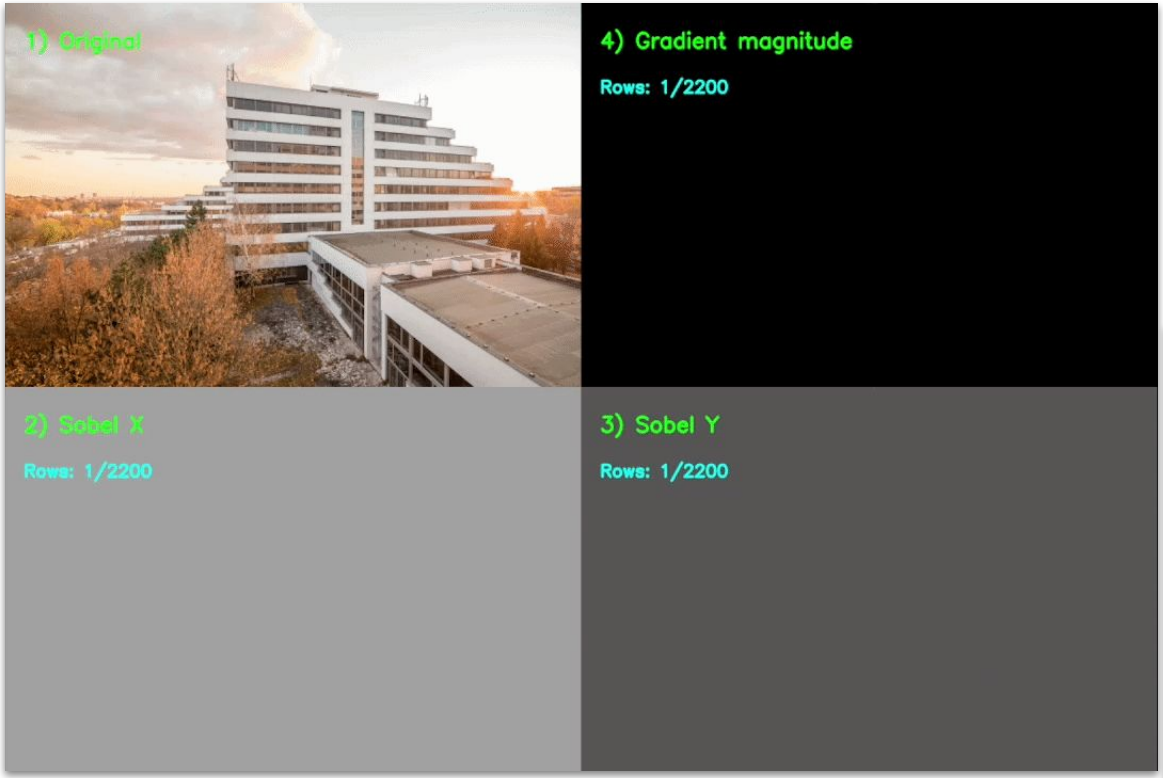
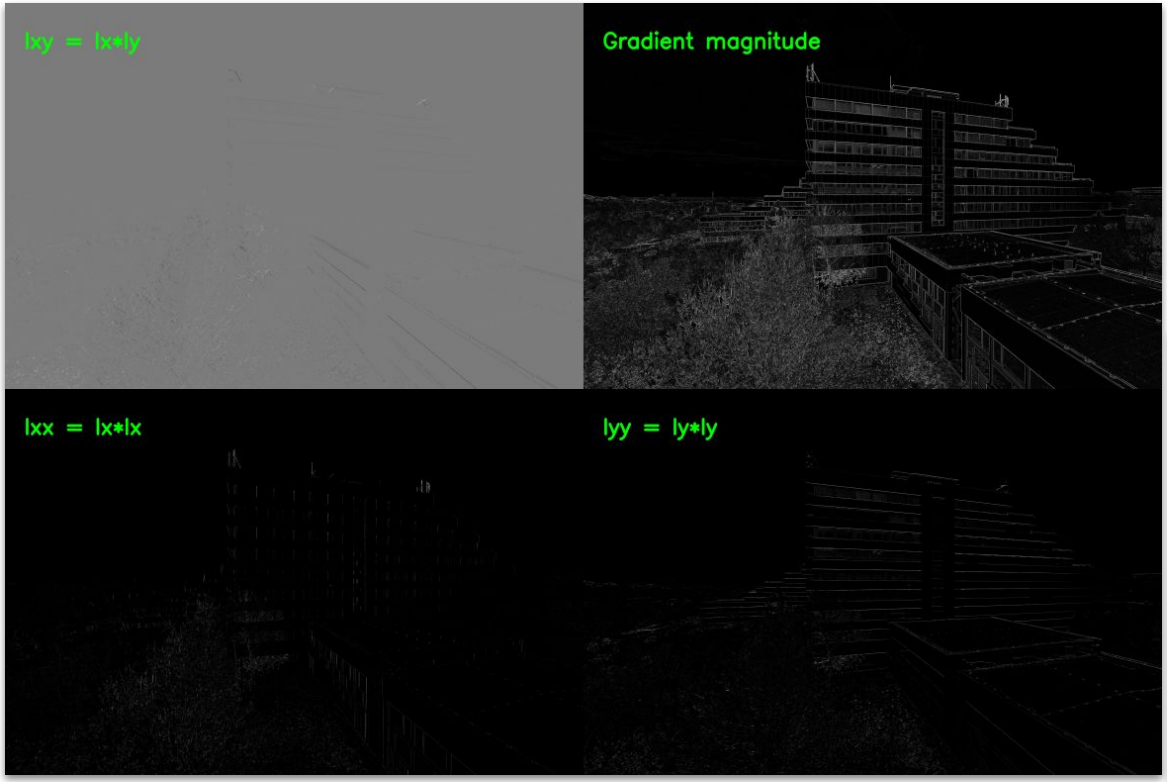
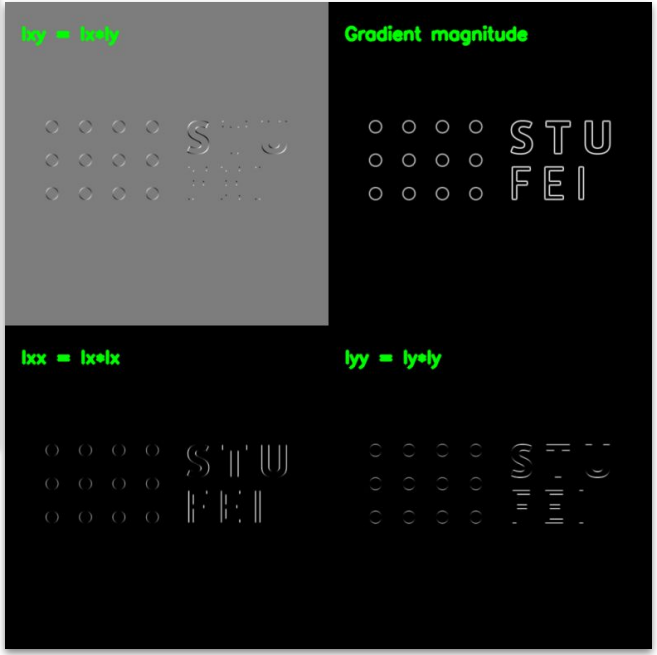



1) Original
4) Gradient magnitude
Rows: 1/2200
2) Sobel X
Rows: 1/2200
3) Sobel Y
Rows: 1/2200

Matica druhých momentov

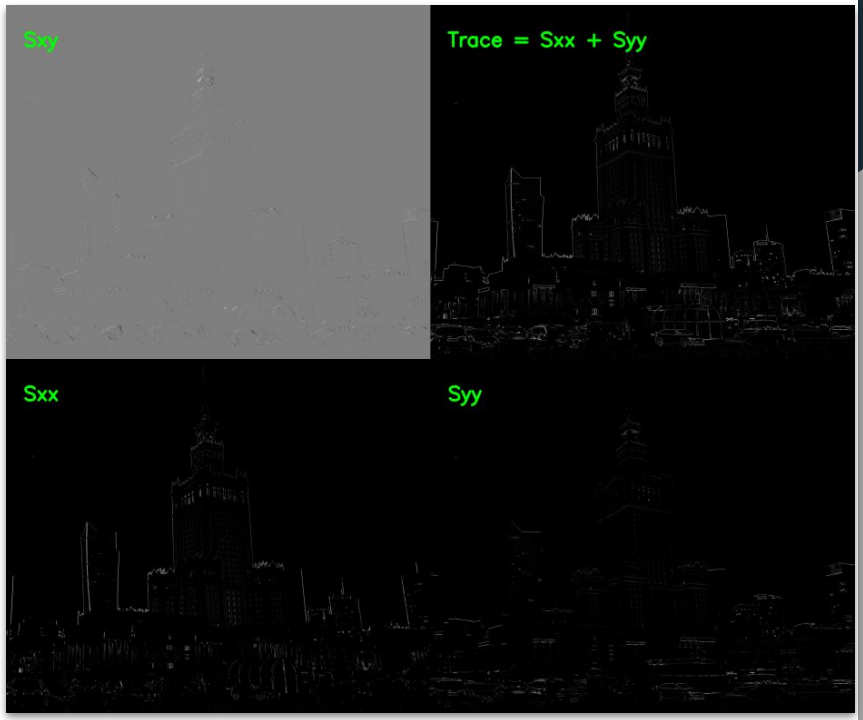
- Produkty gradientov Ixx, Iyy, Ixy
- Štrukturálny tenzor M pre každý pixel
- Po-pixelové násobenie, žiadna konvolúcia

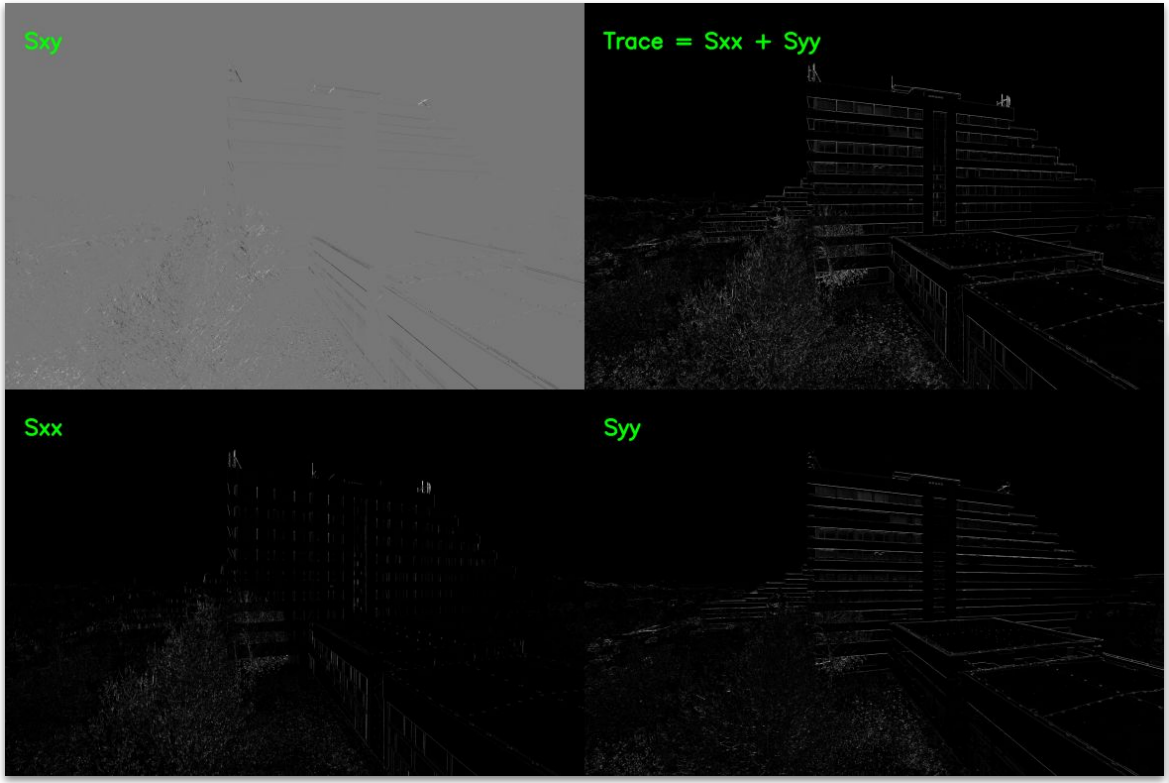

Ixy = Ix*Iy
Gradient magnitude
Ixx = Ix*Ix
Iyy = Iy*Iy

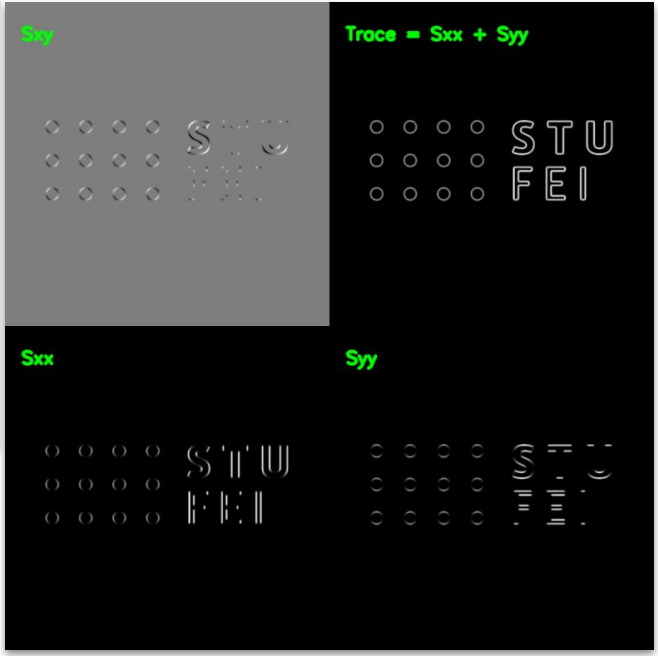

Ixy = Ix*Iy
Gradient magnitude
STU
FEI
Ixx = Ix*Ix
Iyy = Iy*Iy

Gaussovo vyhladenie tenzora

- Gaussovo jadro 3x3
- Vyhladenie produktov Ixx, Iyy, Ixy na Sxx, Syy, Sxy
- Manuálna konvolúcia rovnakou metódou ako v kroku 1
- Bez vyhladenia by bola odozva zašumená




Sxy
Trace = Sxx + Syy
Sxx
Syy


Sxy
Trace = Sxx + Syy
STU
FEI
Sxx
Syy

Harrisova odozva R

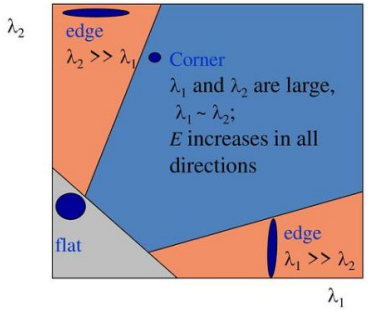
- Pre každý pixel vypočítame odozvu podľa vzorca:
  - $R = \det(M) - k \cdot \mathrm{trace}(M)^2$
  - konštanta k nastavená na 0.04
- Následne prahujeme: ponecháme len pixely kde R > 1 % z maximálnej hodnoty R
- Kladné R znamená roh, záporné hranu, nulové plochu

Measure of corner response:

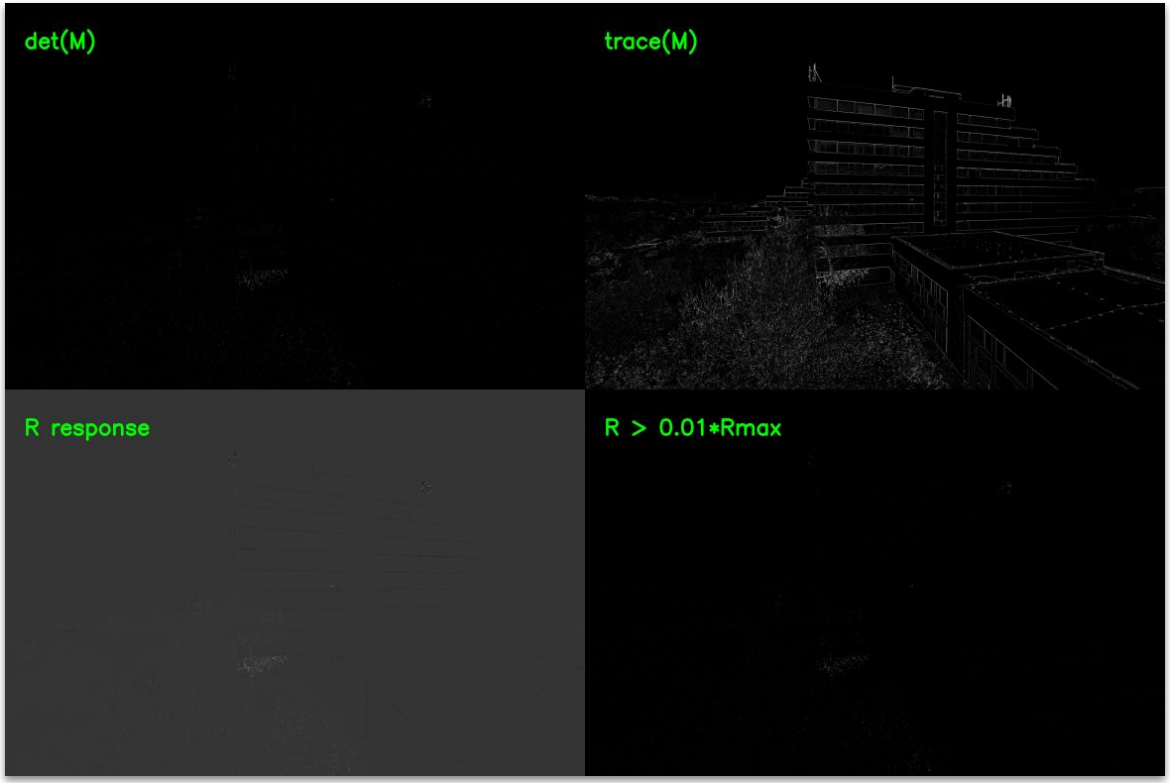
$$R = \det M - k\left(\mathrm{trace}\, M\right)^2$$

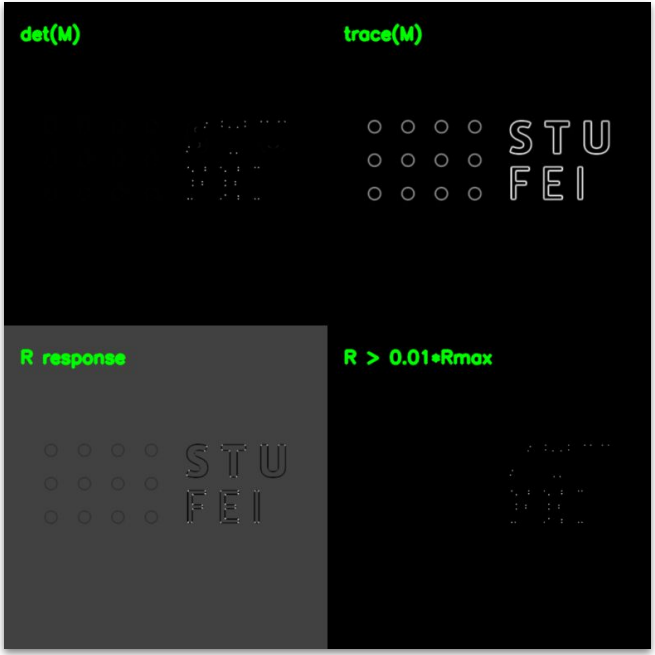
$$\det M = \lambda_1 \lambda_2$$
$$\mathrm{trace}\, M = \lambda_1 + \lambda_2$$

Threshold on value of R



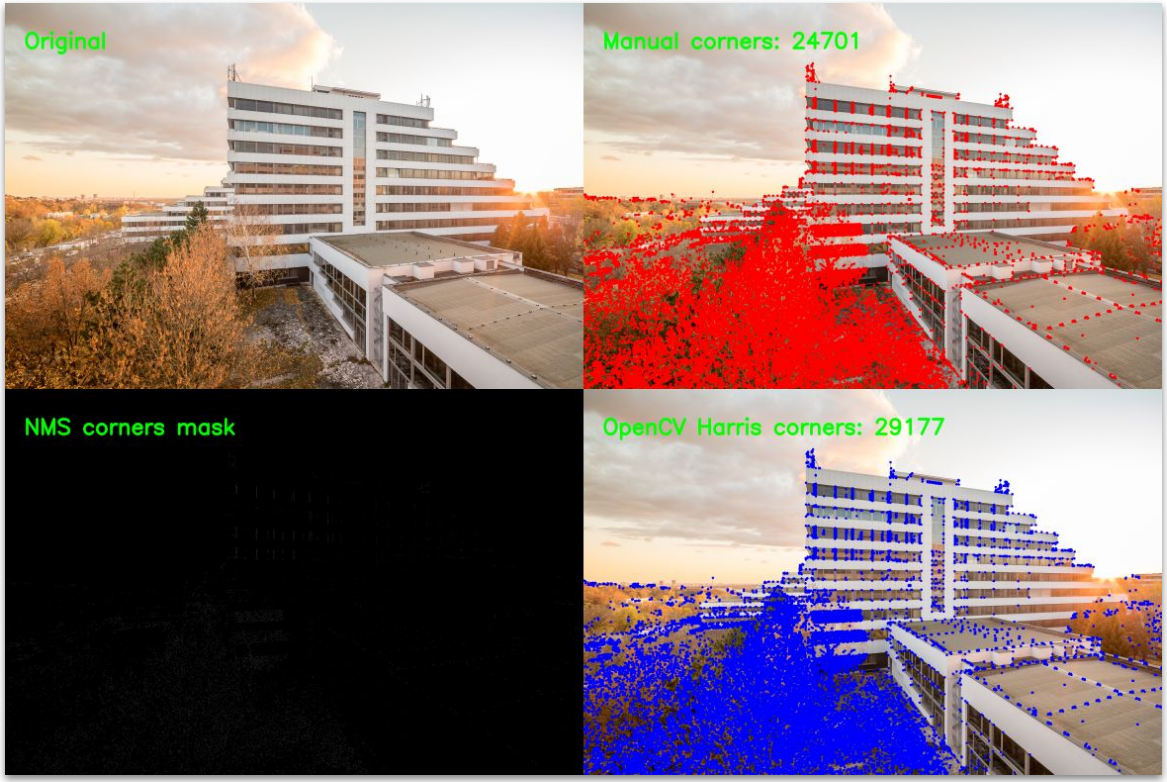
Classification of image points using eigenvalues of *M*:

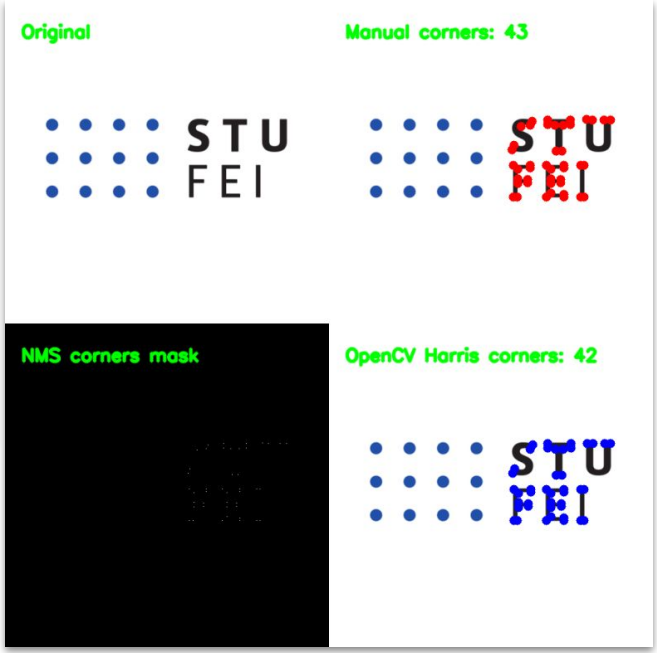

det(M)
trace(M)
R response
R > 0.01*Rmax


det(M)
trace(M)
STU
FEI
R response
R > 0.01*Rmax
STU
FEI

Potlačenie nemaxím (Non-Maximum Suppression)

- NMS v okolí 3x3, ponechanie len lokálneho maxima
- Odstránenie zhlukov detekcií okolo jedného rohu
- Porovnanie s OpenCV cornerHarris s rovnakými parametrami


Original
Manual corners: 24701
NMS corners mask
OpenCV Harris corners: 29177


Original
STU
FEI
Manual corners: 43
STU
FEI
NMS corners mask
OpenCV Harris corners: 42
STU
FEI

Ďakujeme za pozornosť!

Zdroje:

https://datahacker.rs/opencv-harris-corner-detector-part1/
https://datahacker.rs/wp-content/uploads/2019/06/corners.jpg
https://towardsdatascience.com/wp-content/uploads/2020/11/1m9XHMKQPY6mKYsaykuVAsw.png
https://miro.medium.com/0*JUpRtT3SAU1HQmL9.jpg
https://docs.opencv.org/4.x/dc/d0d/tutorial_py_features_harris.html
https://en.wikipedia.org/wiki/Harris_corner_detector
https://www.cs.cmu.edu/~16385/s17/Slides/6.2_Harris_Corner_Detector.pdf
https://thepythoncode.com/media/articles/non-maximum-suppression-using-opencv-in-python/non-max-suppression.webp
https://live.staticflickr.com/65535/32893623777_33403140d3_b.jpg